

Selection constraints of plant adaptation can be relaxed by gene editing

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Climate change has already caused noticeable changes in species-wide traits, such as the well-documented acceleration of spring flowering. Because the evolutionary past has favored certain combinations of traits, some strategies like fast growth with early flowering that are adaptive today are at odds with other plant resilience strategies such as elevated water use efficiency. We know that the evolution of trait combinations is shaped by genomic constraints, but it is unclear whether and how this is affected by natural selection from climate change. Growing hundreds of *Arabidopsis thaliana* natural populations under different rainfall regimes revealed opposing natural selection on flowering time and water use efficiency, with strong antagonistic genetic correlations and contrasting causal alleles identified by Genome-Wide Association analyses. Inactivation of the central flowering regulator *FLC* in multiple wild type accessions, relaxed trait covariance in a genetic background-dependent manner and allowed for the emergence of a novel adaptive trait combination—early flowering and high water use efficiency. Future climates are predicted to escalate conflicts in natural selection among adaptive traits, but our work shows that surprisingly simple genetic changes can partially solve these evolutionary conflicts.

Phenotypic trade-offs across individuals within a species, or across species, are common in nature¹. Trade-offs may arise as a result of local natural selection favoring certain trait combinations^{2,3} or an overlap in the genetic loci involved in those traits, or both. These trade-offs may limit the ability of organisms to simultaneously optimize multiple traits in a new environment, as the most favorable trait combination may not yet exist, which will thus constrain adaptation⁴. For example, plant phenology and ecophysiological traits relevant for adaptation in water limiting conditions—e.g., *escape* from drought periods through rapid reproductive cycling or *avoidance* of water loss through efficient CO₂ uptake and maintenance of internal water pressure—often show strong trade-offs with each other^{5,6}. Given climate change is already reaching ~2°C average warming, and is generating more frequent and extreme droughts⁷, we are observing phenological acceleration across species, including early spring flowering. These phenotypic shifts effectively involve die-off of drought sensitive genotypes⁸ and creates asynchronies with pollinator emergence^{9–12}, suggesting trade-offs in adaptation. We do not yet understand how correlated genomic trait architectures influence how natural selection acts on these phenotype combinations across environments, or whether such genetic trade-offs can be uncoupled. By combining trait data from *Arabidopsis thaliana* populations with field experiments, and using CRISPR/Cas9 to edit a major adaptive locus in multiple populations, we quantify a maladaptive trade-off between early flowering and water use efficiency (WUE), which may impede phenotype evolution due to a genetic constraint. While knocking down a central flowering regulator has the expected negative effect on WUE, it was very surprising that the correlation between flowering time and WUE is relaxed considerably, thus allowing natural selection to act directly to increase WUE.

To understand the phenotypic space of climatic adaptations in *A. thaliana*, we compared 64 ecologically-relevant traits related to life history and drought response in hundreds of geographically diverse accessions (see **Text S1, Fig. S1**). These trait values were compiled from published laboratory and field experiments across subsets of the 1001 Genomes accessions, rather than measured in a single common garden. We determined the main axes of phenotypic variation with Principal Component Analysis (PCA) and overlaid a 2D color scheme in order to differentiate the suites of phenotypes by color (**Fig. 1A**) (**Text SII**). The main PC axis of variation captures the key phenotypic trade-offs of WUE and flowering time (highly correlated traits, $n=248$, Pearson's $r = 0.38$, $p < 6 \times 10^{-10}$) (**Fig. 1B**), stomatal density and stomatal size¹³, high and low seed dormancy¹⁴, and a life history shift between rapid flowering without or only after vernalization¹⁵; while the secondary axis is associated with germination rate (*i.e.* primary dormancy) and root architecture (**Fig. 1A**). The first axis describing within-species variation is meaningful and of general relevance, as it is also the first component structuring life history, lifespan, and physiological variation across hundreds of thousands of vascular plant species¹, often called a “fast versus slow” plant economic spectrum^{16–18}. PC1 also captures a well-characterized continuum in *A. thaliana* between investing resources in early reproduction (escapers) versus prioritizing WUE (avoiders) (**Fig. 1A**)^{5,19–22}. This gradient is apparent in trait correlations, with 24% of all traits involved in escape or avoidance ($n=509$; **Table S1**) being significantly correlated with flowering time (*i.e.*, lifespan)²³, 8% with WUE²⁴, and 15% with growth rate¹⁸ (**Table S3**). This indicates a massive overlap in traits related to adaptation, with substantial potential for shared genetic architectures that could constrain population adaptation phenotypic responses to new climates.

To characterize what traits are adaptive in different climates, we leveraged *A. thaliana*'s broad native range, from Africa to Scandinavia, spanning extreme climate gradients from season lengths to

water availability in the soil. High annual temperature is typically associated with early flowering and thus *escape* (Pearson's $r_{\text{flowering-mean,max,temp.}} = -0.35$, 95% CI -0.41 to -0.30 , $n = 949$, $p = 4.5 \times 10^{-29}$; **Fig. S2**). However, precipitation during summer (Pearson's $r_{\text{fit-bio18}} = -0.20$, 95% CI -0.26 to -0.14 , $n = 916$, $p = 1.1 \times 10^{-9}$; **Fig. 1C**) or during the growing season January-May (Pearson's $r = -0.24$, 95% CI -0.30 to -0.18 , $n = 916$, $p = 2.1 \times 10^{-13}$; **Fig. S2**) correlates with slow growing and late flowering and thus *avoidance*. To help us visualize and generate hypotheses on what climates may drive multivariate phenotype adaptations, we mapped PC phenotypic axes beyond just flowering time across the native geographic range using climate variables and a random forest model (**Fig. 1D**). This directly showed that early and late flowering phenotypes are represented at the extreme latitudes as previously reported (**Fig. 1D**). At low latitudes, with mild and wet Mediterranean winters, populations have evolved to not require cold vernalization cues to flower^{25,26}, and subsequently flower early^{27,28}. This rapid completion of their life cycle is a canonical *escape* strategy to take advantage of a short rainy season. At the high latitudinal extreme of Scandinavia, populations live in climates with many months of snow cover, germinate in the fall, overwinter as small rosettes, and flower late in the season after snowmelt using an *avoidance* strategy. However, this poses the question on why *avoidance* strategies have not evolved in warmer, drought-prone regions, an insight would be important to understand given warming and drying are both expected to increase with climate change⁷.

To test hypotheses on phenotypic selection and adaptation, transplant experiments are the gold standard, as they directly allow comparing the fitness of different populations and their phenotypes in specific environments. Such experiments already conducted in *Arabidopsis thaliana* literature often report natural selection favoring early flowering, low WUE genotypes^{3,20,28-30} (**Table S9**), but in warm and dry climates such as Southern Spain³¹, Eastern Spain²⁹, and Italy³, flowering and fitness are sometimes uncorrelated, and in rare cases is somewhat paradoxically correlated with late flowering. Even in the same location, different precipitation patterns in different years can either favor early- or late-flowering³⁰, suggesting a complex relationship of climate with *escape* and *avoidance* strategies. However, previous transplant experiments often have dealt with regional transplants and panels of *A. thaliana* accessions.

To directly quantify phenotypic selection across trait combinations in precipitation-limited conditions, we combined fitness data from ~500 *A. thaliana* accession grown under rainfall-manipulated common garden experiments in Spain and Germany with our curated and imputed trait database (**Fig. 1E**³²). The well-documented selective pressure favoring early flowering time^{20,30} in annual plants is immediately apparent ($s_{\text{fitness}} = -0.191$, $p_{\text{boot}} < 1 \times 10^{-3}$; **Table S7**). Partitioning fitness data into survival to reproduction and fecundity revealed a key incongruence in the hot and dry Spain environment: early flowering was favored by selection for survival (**Fig. 1F**) ($s_{\text{survival}} = -0.282$, $p_{\text{boot}} < 1 \times 10^{-3}$; **Table S7**), while later flowering was favored for fecundity (**Fig. 1G**) ($s_{\text{fecundity}} = 0.085$, $p_{\text{boot}} < 5 \times 10^{-2}$; **Table S7**). This trade-off in the direction of selection is not limited to flowering time and is also observed for other traits contributing to PC 1, including growth rate (**Table S9**) and primary seed dormancy (**Table S10**). PC 1 itself also shows opposing selection gradients depending on the fitness component considered ($s_{\text{survival}} = -0.239$, $p_{\text{boot}} < 1 \times 10^{-3}$; $s_{\text{fecundity}} = 0.109$, $p_{\text{boot}} < 5 \times 10^{-2}$; **Table S6**). This is consistent with delayed flowering permitting plants to accumulate more resources for reproduction but at an increased risk of mortality³³, however it could also indicate there are indirect selective pressures for flowering time influencing trait evolution³⁴.

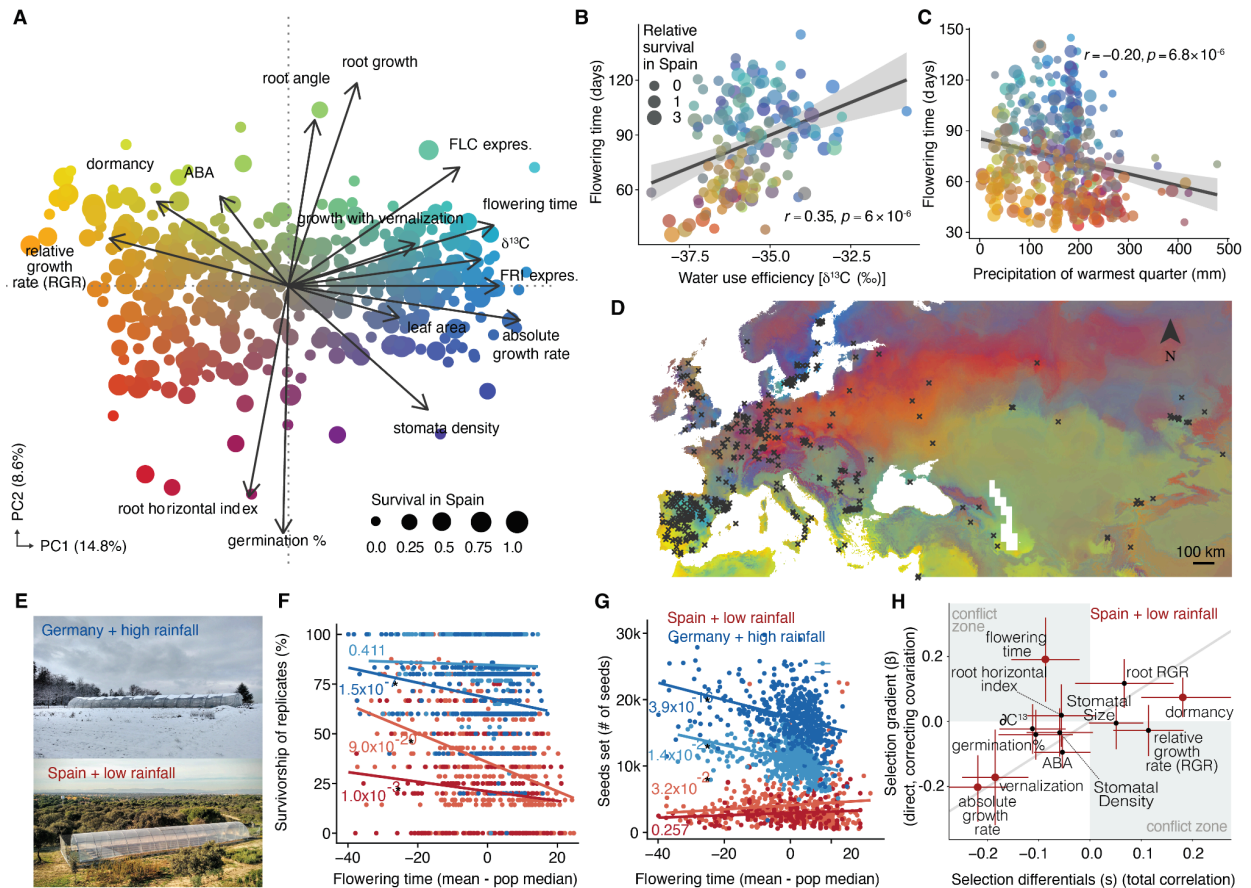


Fig. 1 | Escape and avoidance strategies across the *A. thaliana* native range leads to natural selection conflict.

(A) Principal Component Analysis (PCA) of 64 ecophysiological and life history traits in 515 accessions (arrows for 15/64 loadings indicated for visual clarity). Circle size indicates fitness in a Spanish field experiment³² and a color gradient is mapped on the points to distinguish phenotype groups. (B) Correlation between non-imputed data for flowering time at 16°C and WUE (using $\delta^{13}\text{C}$ as proxy); colors correspond to those in (A). The correlation coefficient is reported for all accessions with trait data ($n = 248$), while the plot shows the subset with fitness data ($n = 160$ of 515 total accessions). (C) Correlations between non-imputed data for flowering time at 16°C and precipitation of the warmest quarter for 485/515 accessions (Text SII); colors correspond to those in (A); trendlines show fitted linear models for visualization. (D) Predicted phenotypes in color scale from (A) across the Eurasian native range of *A. thaliana* (Text SII). (E) Common garden sites. The absolute correlation between flowering time at 16°C (median-centered) and (F) survival as the proportion of surviving-to-reproduction replicates and (G) total number of seeds set³². Associated p values (linear model) are in the same color as trend lines, with darker lines indicating low and lighter lines high planting density; note these are not selection differential p values but do show the same patterns; trendlines show fitted linear models for visualization. (H) Total selection differentials (s) vs. direct selection gradients (β , after accounting for trait covariation), using survival data from the hot and dry common garden experiment (Spanish field site), with low planting density. Large red circles indicate significant β and s ($p < 0.05$), small black circles indicate n.s., circle size reflects only the binary significance threshold, not the magnitude of p -values; error bars from 100 bootstrap replicate estimates (Text SIII); gray background areas indicate conflict zones where the signs (+/-) of total and direct selection conflict and are in the opposite direction.

Because PC1 includes several key traits underlying the trade-off between *escape* and *avoidance*, we wanted to disentangle phenotypic selection pressures by accounting for correlated traits in a multivariate selection analysis³⁵. Specifically, we aimed to quantify *direct* effects of natural selection in each of 12 key focal traits excluding the highest collinear traits (**Fig. S3, Text SII**). Most traits remained consistent in their direct, correlation-free selection gradient estimates, compared to the total selection differentials. For instance, for absolute growth rate, the total selection estimate ($s_{\text{survival}} = -0.219$, $p_{\text{boot}} < 10^{-3}$) was very similar to the direct selection estimate ($\beta_{\text{survival}} = -0.206$, $p_{\text{boot}} < 10^{-3}$; **Fig. 1H**), indicating that there is no influence of correlated trait variation on selection for absolute growth rate. Likewise, the vernalization requirement for flowering time ('vernalization') and primary dormancy ('dormancy') also show consistency between total and direct selection estimates (**Fig. 1H**). The same relationship between direct and total selection for these traits is observed in Germany, but the magnitude of the effect is much smaller, reflecting weaker selection pressure under the less stressful climate (**Fig. S5**). Strikingly, however, the sign of selection for flowering time became reversed from negative to positive (**Fig. 1H**). This was not due to the effect of fecundity component on overall fitness (**Fig. 1G**), but we found the sign flip in survival fitness component, suggesting an advantage of late flowering accessions (*avoiders*) once we had accounted for trait correlations ($s_{\text{survival}} = -0.087$, $p_{\text{boot}} < 10^{-2}$; $\beta_{\text{survival}} = 0.190$, $p_{\text{boot}} < 10^{-3}$; **Fig. 1H; Table S13**) (leave-one-out re-analyses indicate this reversal can be seen when accounting for growth rate and vernalization see **Text SIII, Table S13**). This is the first positive selection gradient detected between flowering time and fitness in a warm, water-limited common garden environment.

A fitness advantage of late-flowering accessions of *A. thaliana* in hot and dry environments would not be completely unexpected, as later flowering plants are typically more resilient to water limitation^{20,36}. Late flowering is correlated with water accumulation through alterations in root development, stomata control, and overall higher WUE (**Fig. 1A**). However, we would thus predict that WUE is also directly favored in such environments. Instead, we found that total selection differential was negative ($s_{\text{survival}} = -0.112$, $p_{\text{boot}} < 10^{-3}$) and the direct effect was non-significant ($\beta_{\text{survival}} = -0.023$, ns; **Fig. 1H; Table S13**). The only significantly positive total selection differential of WUE was on fecundity ($s = 0.048$, $p_{\text{boot}} < 0.01$; **Table S13**). While sampling and environmental noise can produce slight differences in phenotypic estimates, direct selection being so strongly inconsistent with total selection is an uncommon observation relative to many plant and animal species and thousands of measured morphological, phenological, or physiological selection estimates^{34,37}. However, given flowering time is such a fundamental trait in trade-off with other water use traits, this conflict of selection may have broad implications in impeding multivariate trait evolution in plants.

We then asked what are the evolutionary response consequences of this total vs direct natural selection conflict, in the light of trait heritability and genetic correlations^{4,38,39}. To answer this question, we first performed systematic genome-wide association (GWA) analyses with the accessions from the 1001 Genomes Project²³ for each of the 12 key traits⁴⁰, including pairwise multivariate linear mixed model (mvLMM) GWA⁴¹ to estimate narrow sense heritability and genetic correlations. Most traits had moderate-to-high SNP-based heritability ($h^2 = 0.53$ [IQR 0.29-0.79], $n = 1,848$) and were polygenic ($\gamma n = 48.5$ causal loci [IQR: 17.2-68.1]) (see all values in **Table S15**, and distributions **Fig. S12**). The strongest, most consistent genetic correlation was between flowering time and growth rate ($r_{\text{g-flowering-growth}} = 0.68$, 95% CI 0.57–0.79; **Fig. S16**), and both were in turn significantly correlated with WUE at the genetic level ($r_{\text{g-flowering-WUE}} = 0.45$, 95% CI: 0.35–0.56, **Fig. 2A**; $r_{\text{g-growth-WUE}} = 0.44$, 95% CI 0.33–0.55). To solidify the

inference that genetic correlations may limit adaptive paths, we applied the Breeder's equation, which accounts for heritability and genetic correlations amongst our focal 12 traits, to predict the average phenotype change of each trait (*sensu* Lande & Arnold, ³⁵; **Fig. 2D**). Only traits with consistent total (*s*) and direct (β) selection estimates, as well as high heritability and aligned genetic covariances, would evolve in the expected direction of natural selection. One such trait was growth rate, which showed significant selection towards low absolute biomass growth—consistent with the small plant, fast-cycling escape strategy — and we predicted a significant decrease in the mean trait value in the next generation (**Fig. 1H**). Consistent with the fact that total differential and direct gradients of selection were variable in flowering time and WUE, and their genetic correlations, we found no significant shift expected in the mean trait value of neither flowering time nor WUE (**Fig. 2D, Table S21**) despite both having significant total phenotypic selection (**Fig. 1H , Table S13-14**). Rather, we predicted the counter-intuitive “evolutionary stasis” of these traits in a Mediterranean climate with water limitation ($\Delta z_{\text{flowering}}=0.136, p > 0.05$; $\Delta z_{\text{WUE}}=-0.001, p > 0.05$, **Fig. 2D, Table S21**). This can be interpreted as evolvability of the trait does not equate to heritability and selection, but is limited by the genetic covariance among traits, G-matrix, which dictates the paths of least resistance in evolution (**Table S21**) ⁴². In conclusion, though the estimate for flowering time and water use efficiency is leaning positive in water-limiting environments, the influence from correlated trait variation may deflect responses to selection from the expected adaptive evolution paths. As it is expected that the climate in much of Europe will become more similar to the Spanish hot/dry environment, such conflicting evolution may be exacerbated (**Fig. 2E, F**).

We therefore wanted to understand what is the root cause of this evolutionary constraint, and asked if there are pleiotropic gene variants across traits. We then examined the effect sizes on flowering time and WUE GWA analyses, specifically studying the top 0.05% SNPs (*n*=274) in our mvLMM GWA. This showed a striking parallel; alleles increasing WUE also increased flowering time, and vice versa (Fisher's Exact Test odds ratio = 3,618, $p < 2.2 \times 10^{-16}$; **Fig. 2B**). The expected allele frequency changes (**Fig. 2C**) show a static trend where putatively pleiotropic loci impacting both traits sometimes increase and sometimes decrease in frequency implying evolutionary stasis. Co-locating SNPs with gene annotations, we found many of the major effect alleles are in well-known flowering-time genes, including *FLC* and *SVP* (**Fig. 2A**), *DOG1*, and *VIN3* ¹⁵, or *FRI* ⁴³, known to have pleiotropic effects in flowering time and WUE²⁰ (**Fig. S13**). Intriguingly, we also discovered a highly significant locus on chromosome 1 so far undescribed, as appearing consistently in our different analyses (**Text SIV**). The top SNPs span a region of ~7.5 kb and 5 candidate genes (**Fig. S18; Text SIV**), with the expression of three of these genes correlating with flowering time ⁴⁴, and another with WUE (**Fig. S19**). To estimate the effect of natural selection on these alleles rather than in phenotypic traits, we conducted an additional GWA with fitness from Spain hot/dry conditions ³². The chromosome 1 alleles that increased both flowering time and WUE (**Fig. 2B**) were negatively correlated with fitness (mean effect size = -0.054 [0.10]; **Fig. 2A**). After including growth rate as cofactor though, their correlation with fitness reversed (mean effect size = 0.173 [0.190]; **Fig. 2A**), meaning after the negative selective effect from growth rate was accounted for, the remaining variation in fitness suggests positive selection for these alleles. These same alleles also correlated negatively with survival (mean effect size= -0.117 [0.089]) but positively with fecundity (mean effect size= 0.089[0.070]). We speculate that this locus is pleiotropic and subject to antagonistic selection, recapitulating the selection constraint originally observed at the trait level between flowering time and growth rate.

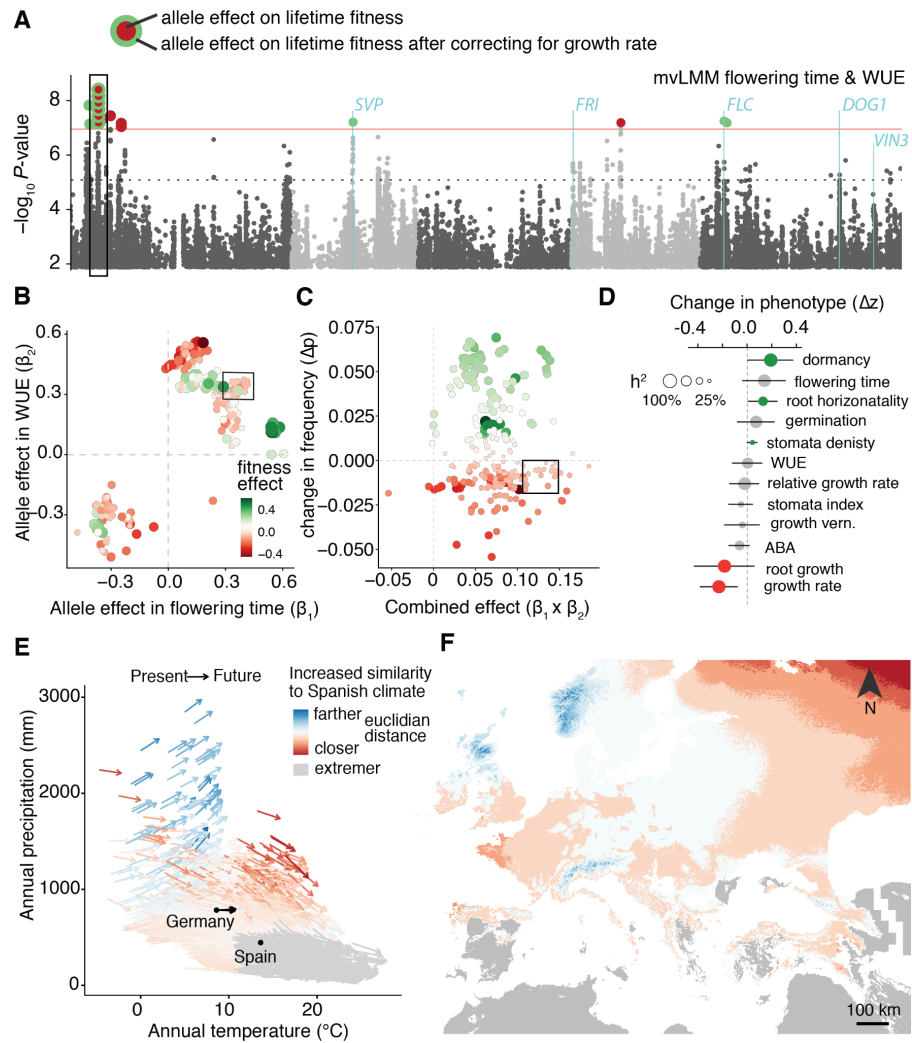


Fig. 2 | The joint genetic architecture of escape and avoidance constrains trait evolution in water-limiting environments.

(A) Manhattan plot of mvLMM GWA of flowering time and WUE (δC^{13}) (n=248). Red line is the Bonferroni-corrected genome-wide significance threshold of $-\log_{10} P\text{-value} > 7.03$, black dashed line indicates the top 0.05% of alleles with $-\log_{10} P\text{-value} > 5.13$; annotated genes are significant for flowering time¹⁵ and are identified in our extended analysis of these data (Text SIV). Significantly associated SNPs are colored based on the effect size of a separate GWA with lifetime fitness from the hot, dry Spanish environment, where allelic effects on fitness were estimated; circle color indicates alleles with a positive (green) or negative (red) association with fitness; outer circle color of alleles on chromosome 1 indicate the effect each allele has on fitness after correcting for growth rate. (B) Top 0.05% SNP effect sizes for flowering time (β_1) and WUE (β_2) from GWA described in (A), circles colored and sized by lifetime fitness effects estimated from fitness GWA described above; black box highlights the same alleles as in (A). (C) Expected allele frequency change in response to selection (Text SIV) by the combined allelic effects for flowering time and WUE; black box highlights the same alleles as in (A) and (B). (D) Predicted mean phenotypic response to selection across traits, size of circles indicates SNP-based narrow-sense heritability estimate represented as a % (see Table S15), color of the circles indicates positive (green), negative (red), or no (gray) predicted change in the mean trait value of the next generation. (E) Predicted change in annual precipitation and temperature similarity to the Spanish field site climate in 2050 (measured as Euclidean distances). (F) Map of climate trajectories in (E). Climate maps were retrieved from Max Planck CMIP5 IPCC projections of

business-as-usual scenarios (worldclim.org). Gray areas indicate regions where temperature will increase while precipitation will decrease below the average precipitation at the Spanish field site (*i.e.* more extreme conditions).

Predictions of evolvability based on quantitative genetics, which assumes polygenic effects underlying traits, may be insufficient to understand evolutionary constraints. We then wondered whether, in contrast, a large-effect allele with high trait penetrance, could bypass these constraints, shifting trait variation despite the persistent polygenic covariance structure³⁵. To test this expectation directly, we experimentally materialized a thought experiment of a large-effect allele accelerating flowering time, that becomes fixed in a population through CRISPR/Cas9 engineering. We asked how this would impact trait correlations with WUE and whether it would release indirect natural selection constraints, permitting evolution of WUE traits. We used *FLOWERING LOCUS C (FLC)* as a proof-of-principle large-effect gene, because *FLC* encodes a master regulator that prevents premature flowering in winter and is well understood. Our GWA analyses had identified natural *FLC* alleles as controlling the bulk of flowering time variation (**Fig. S13**), and the mvLMM GWA identified a statistically significant peak at the *FLC* locus, indicating this gene also may play a pleiotropic role in the shared genetic variation between flowering time and WUE (**Fig. 2A, Table S31**). In natural populations, *flc*-knock-out (KO) and severe knock-down (KD) mutations nearly always result in an early spring-flowering phenotype and segregate in warm environments at the Southern extent of the range^{26,45}. It is therefore theoretically possible that a *flc*-KO mutation could become prevalent in warm environment populations and accelerate flowering time for an entire population.

To engineer *flc*-KO mutations, we designed gRNAs in two downstream and two upstream target regions of *FLC* (**Fig. 3A**) and transformed a diverse set of 62 wild type *A. thaliana* accessions. Overall, 84% of the edited lines had reduced *FLC* transcript levels, with 39 likely having large deletions (KOs) and the remainder having weaker mutations (KDs) (**Fig. 3A**). As flowering time is correlated with WUE, we expect modulation of *FLC* activity to have a pleiotropic effect on WUE. We measured flowering time and δC^{13} for a subset of the *flc* lines (n=29) and their wild-type original counterparts and reference Col-0 accession (n=20, due to tissue limitation) (**Fig. 3B**). In most *flc* lines, flowering was accelerated (mean [se] = 22.1 days [1.0], n=29), compared to wild-type parents (73.3 days [6.7], n=19) (**Fig. 3B**). There was also a significant reduction of WUE (mean [se] = -30.81‰ [0.082], n=29, compared to wild-type of -30.24‰ [0.095], n=20; *t* test, Benjamini-Hochberg correction, $p_{adj.} < 0.05$; **Fig. 3B**). Remarkably, the relationship between flowering time and WUE while significant was flatter amongst the all-early-flowering-*flc* lines compared to the wild types (**Text SVIII**), evidence for a relaxation of phenotypic genetic constraints in the *flc* population (**Fig. 3B**), and theoretically enabling natural selection to act on WUE.

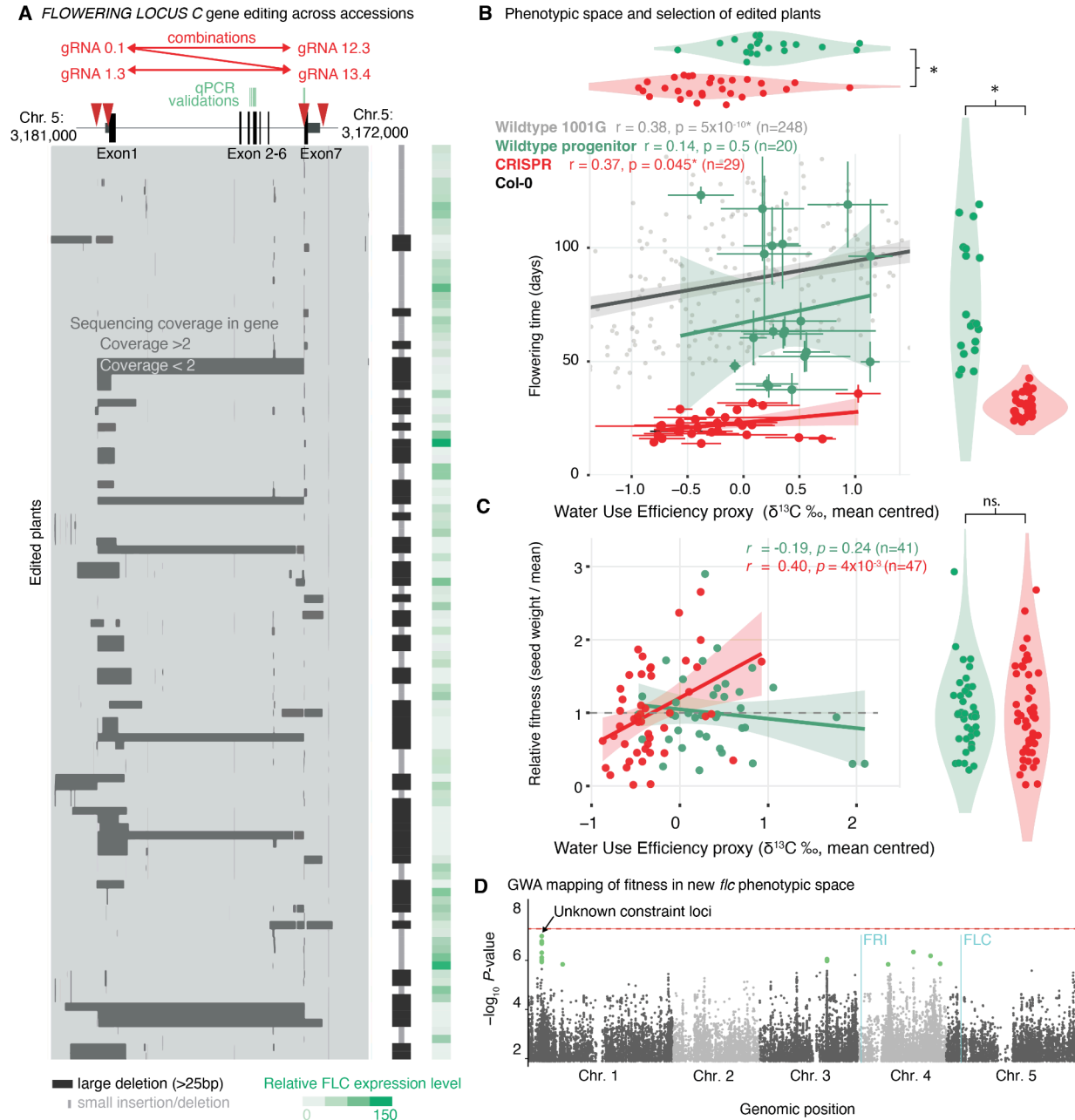


Fig. 3 | CRISPR/Cas-mediated gene editing of diverse *A. thaliana* accessions relaxes trait correlations between flowering and WUE that can become unfavorable in future.

(A) Top, diagram of the *FLC* locus. Black boxes represent exons 1 - 7, gRNA target sites indicated by red arrows. The red connectors indicate gRNA combinations in single editing constructs. Bottom, each row represents one *flc* mutant line ($n=112$ lines of $n=62$ wild type accessions, median ~ 2 lines successfully transformed per accession). Grey scale corresponds to whole-genome sequencing coverage at the *FLC* locus in different lines. Dark gray signifies reduced coverage indicative of deletions. On the right, membership of each line in a “large deletion” and “small mutation” class is indicated. On the far right, relative *FLC* expression in nine-day-old plants grown in long days is shown. (B) Correlation of flowering time and $\delta^{13}\text{C}$ for *flc* mutants and wild types in 20-22°C under LD. Red indicates *flc* mutants (29 mutants / 62 accessions with successful phenotyping) and green wild types (19/62 accessions with successful phenotyping). WUE proxy of $\delta^{13}\text{C}$ is centered in the mean of each experiment for

comparability (* indicates one-sided Mann-Whitney U rank test significance $p < 0.05$) (C) Correlation of seed weight and WUE for *flc* mutants and wild types in warm conditions without water limitation (47 *flc* lines and 41 progenitors with successful phenotyping out of 62 wildtype targeted accessions). WUE proxy of $\delta^{13}\text{C}$ is centered in the mean of each experiment for comparability. (D) Manhattan plot of mvLMM GWA of flowering time and WUE ($\delta^{13}\text{C}$, mean centered) for the *flc* lines, with points colored by selective effect on alleles measured by univariate GWA using fitness of *flc* lines in warm conditions without water limitation.

To test whether the change in trait covariance could change natural selection on flowering time and WUE, we grew 47 *flc* lines and their wild-type progenitors ($n = 41$) in warm conditions without water limitations (see Methods). While in the growth chamber we could not see any survivorship differences due to the milder conditions than field setups, we used seed weight as fecundity fitness proxy, and now found a positive relationship of fitness with WUE in the *flc* lines ($r = 0.4$, $p = 4 \times 10^{-3}$), in contrast to the negative but non-significant trend in the wild-type progenitors ($r = -0.19$, $p = 0.24$, $n = 41$) (Fig. 3C). Consistently, the relationship between fitness and flowering time trended toward favoring early flowering in the wild types but later flowering in the *flc* lines (Fisher's $z = 2.07$, $p = 0.038$), although both flowering trends were weaker and non-significant (Text SVIII.2). Some accessions even had the unique combination of early flowering and high WUE phenotypes. These results are significantly different from a longstanding observation in annual plants like *A. thaliana*, where positive phenotypic selection on WUE has not been reported; instead, studies have generally found selection favoring low water-use efficiency, even in water-limiting conditions²⁰. We speculate that it may be the strong effect of life history strategy in fitness that may have hindered the observation that high-WUE accessions do in fact perform better under hot, high evapotranspiration conditions. We finally performed a mvLMM GWA on these 47 *flc* mutant lines using the flowering time and WUE trait data and assessed the selective effects on alleles using the fitness data from the experiment. We found that the alleles on chromosome 1 that contribute to increased WUE (Fig. 3D) are now under positive selection (mean effect size = 0.0082 [0.002]). These results support the idea that some large-effect loci, despite their pleiotropic effects, may change the landscape of multi-trait correlations enabling the evolution of novel trait combinations beyond what we had theoretically predicted.

Conclusion

Future climate change is expected to generate novel environments to which species have not yet been exposed in the past and may thus not have evolved the optimal phenotypic combinations. Using gene editing, we have simulated a possible future population that had a large-effect loss of function mutation become fixed, drastically impacting phenotype distributions and the direction of selection. Phenotypic variation within *A. thaliana*, and likely other species, is shaped by genetic variation and past selection forces that have generated an *escape-avoidance* strategy continuum. Given that rapid climate change has already led to early flowering in many species^{9–11}, we expect natural selection to act along “axes of least resistance”, that is the most easily accessible portions of phenotypic space due to high heritability and low polygenicity, even if other beneficial strategies are theoretically possible. Although predicting future evolutionary trajectories is complex, we speculate that large-effect alleles affecting flowering time exist in natural populations, and that these will rapidly increase in frequency due to natural selection in a hotter-and-drier native climate. After the first response to selection, we predict increased WUE will be selected for. However, the short-term evolution of early flowering time will have a detrimental pleiotropic effect in WUE, the consequences of which for adaptation are so far unknown. The expanding genomic

and molecular biology knowledge across plant species and democratization of gene editing technologies should enable us to disentangle complex trait correlations, which is an underestimated constraint to evolution that could lead species to local minima in the fitness landscape that are difficult to escape.

Materials and Methods

Trait data curation

The database of 1,862 traits was assembled from 108 published sources encompassing a combination of laboratory and field experiments that use a subset of *A. thaliana* accessions from the 1001 Genomes Project (**Table S1; Text S1**). The trait data are averages for a genotype across replicates measured in replicated environments and were not measured in a single experiment or for single plants. We classified traits into a general functional category, with 1,282 traits as being related to one of three drought adaptation strategies; *escape* (n=127), *avoidance* (n=382), and *tolerance* (n=773) (**Fig. S1**). Because most traits had only been measured on subsets of 1001 Genomes accessions, with subsets overlapping to different degrees (**Fig. S1**), we used the R package *missForest* to fully impute the trait dataset (see **Text S1; Table S2**).

Trait associations

For PCA, traits related to seed dormancy, vernalization, germination, flowering time, leaf traits, roots, stomata, growth rate, and stress response were isolated (n=205). Highly correlated traits were removed ($R^2 > 0.7$), resulting in a total of 64 usable traits (**Fig. S1**). We estimated how many of the 1,862 traits were correlated with four flowering time datasets, WUE, and growth rate, using Pearson's correlation coefficient with a significance threshold of 0.05 and the non-imputed trait dataset (**Table S3**). We also downloaded bioclimatic, temperature, precipitation, and evapotranspiration rate estimates from WORLDCLIM 2.0 for all localities associated with the 515 accessions for which fitness data were available, and estimated Pearson's correlation coefficient between climate data and these six target traits (see **Text SII; Table S4**).

Selection estimates

We computed total selection differentials s (ref. ³⁵) for all non-fitness traits (**Table S5**) and for 20 PC axes explaining 75% of trait variation (**Table S6**), highlighting focal traits such as flowering time (**Tables S7, S8**), growth rate (**Table S9**), seed dormancy (**Table S10**), and WUE (**Table S11**). Total selection was calculated as the covariation between relative lifetime fitness w and a given trait z (mean centered and variance scaled): $w = sz$, where s is the total selection differential. Fitness data were from an outdoor common garden experiment in two different environments, Madrid, Spain (m) and Tübingen, Germany (t) (**Figure 1E**) ³². Trait data were mean-centered and variance scaled prior to estimating s . Additionally, s was estimated from 100 bootstrap replicates with replacement in order to generate confidence intervals and determine significance.

To understand the direct effects of natural selection on traits, we accounted for correlations amongst traits using the multivariate selection approach, $\beta = P^{-1}s$, where P is the phenotypic variance-covariance matrix ³⁵, s is the previously quantified total selection differential (without accounting for correlations), and β is the gradient of selection (or direct selection) that captures the independent effects of each trait that cannot be attributed to their associations with other traits. Through a series of trait decorrelation, and

expert selection, we narrowed the analyses to 12 focal traits related to both seasonal and drought adaptation (**Table S12, Fig. S3**). We focused this analysis by only using fitness data from the low water Spanish site (hot and dry; **Fig. 1H, Table S13**) and the high water German site (cool and wet; **Fig. S4, S5 Table S14**) (see **Text SIII**). As selection analyses require complete trait matrices, we performed these analyses using imputed trait values. We note that while imputation can slightly alter trait distributions (**Fig. S3**), the overall patterns of selection were qualitatively robust (see **Text SIII.2**).

Genome wide associations

To perform genome wide association (GWA) analyses, we started with the 1,135 accessions and 11,769,920 SNPs from the 1001 Genomes Project (ref. ²³). Depending on trait data coverage, not all samples and SNPs were used as we ran GWA on first the focal 12 traits, but also on all traits (n=1,862) using the raw data, quantile-normalized data, and imputed data. For each trait we conducted GWA using a univariate linear mixed model (LMM) implemented in GEMMA (ref. ⁴⁰). This model corrects for genome-wide background differentiation collinear to the phenotype of study (i.e. relatedness and population structure correction). No minor allele frequency (MAF) cutoff was used unless stated (**Text SV**). Additionally, using the original imputed SNP matrix from <http://arapheno.1001genomes.org>, which contains 10,709,466 SNPs, we used a subset of 1,353,386 SNPs for all 1,135 1001 Genomes accessions in a Bayesian sparse linear mixed model (BSLMM) GWA, also implemented in GEMMA (ref. ⁴⁶), for each of the 1,862 phenotypes (see **Text SVI**). For the 12 target traits used in the selection analysis (**Table S16**), pairwise multivariate LMM (mvLMM) GWA were performed ⁴¹. Again, analyses were run with non-imputed and imputed data, and when the data were not imputed only 16/66 pairs of mvLMM GWA were possible given lack of overlapping samples and large number of parameters to estimate (**Text SIV**).

Fitness GWA

Fitness data from field experiments of 515 accessions ³² were used in a univariate LMM GWA with no MAF cutoff and corrected for population structure using a kinship matrix. The sign of the effect size estimates for each allele indicated whether the allele has a positive or negative effect on the quantitative fitness measure. We use these effect size estimates to color alleles to indicate positive (green) and negative (red) selection (**Fig. 2A, B, C**). To account for the variation in fitness that is explained by absolute growth rate variation, we estimated a linear model of fitness explained by growth rate and then used the residuals from this model in an additional LMM GWA to see the remaining selective effect on alleles (**Fig. 2A**).

Prediction of phenotypic response to selection

We inferred the population responses to this natural selection by using the Breeder's Equation or selection response equation (ref. ³⁶), $\Delta z = GP^{-1}s$, where G is the additive genetic variance-covariance matrix constructed from heritability and genetic correlation estimates among 12 traits, P is the phenotypic variance-covariance matrix, and s is the raw covariance of the trait with lifetime fitness in hot-dry conditions. For this, we used the estimates of s from lifetime fitness in all environments from ref. ³², but primarily report the hot-dry environment (Madrid (m) and low water (l)), but also mention results for WUE in the cool-dry environment (Tübingen (t) and low water (l)). For the genetic variance-covariance matrix, we used the SNP-based heritability estimates of target traits using the GWA results (**Table S16**), and the genetic correlations estimated from the mvLMM GWA from imputed data (**Fig. S17**). An alternative approach to Breeder's equation is the Hansen's evolvability approach (see **Table S28**) ^{42,47}.

Climate projections

We downloaded the averaged historical (1970-2000) bioclimatic variables⁴⁸ for the Eurasian range of *A. thaliana* at 5 km² (2.5 minutes) resolution. Future predicted climate data were retrieved from the Max Planck CMIP5 IPCC model of a business-as-usual scenario⁴⁸. We plotted the 1970-2000 average value of annual precipitation and mean annual temperature as the beginning of the arrows (**Fig. 2E**) and the 2050 predicted values as the end of the arrows to indicate the trajectory of the climate in precipitation and temperature space. Arrows indicate whether the euclidean distance (of both temperature and precipitation) between the current and 2050 climates becomes more similar to the current Spanish climate (red) or more dissimilar (blue) (**Fig. 2E**). Trajectories with predicted precipitation and temperature more extreme than the Spanish site are in gray. The intensity of the trajectory was then mapped back to the initial localities across the Eurasian range to visualize where in the native range it will become more like the Spanish or German climate (**Fig. 2F**).

CRISPR/Cas9 gene editing of FLC

We selected 60 late- and two early-flowering accessions of *A. thaliana* from the 1001 Genomes Project²³, that represent 13 out of 25 of the major FLC haplotypes (**Fig. S21B** and **Table S23**), for CRISPR/Cas9 gene editing. gRNAs were designed using CCTop and no off-target binding was predicted⁴⁹. For CRISPR/Cas9 mutagenesis, we used a slightly modified previously established toolset⁵⁰ with a supermodule destination binary vector carrying a plant-codon optimized Cas9 driven by a *UBQ10* promoter (**Text SVI**). We additionally mutated a TTTT stretch to TTCT by introducing a unique XbaI site. Constructs expressing pairs of four gRNAs (0.1 and 13.4, 1.3 and 13.4, 12.3 and 0.1; **Fig. S21D**) were transformed as pools into the *Agrobacterium tumefaciens* strain ASE⁵¹. Accessions were transformed with the floral dip method⁵² (**Text SVI**).

Mutant screen

T₁ seeds from all pots per accession were pooled and screened for seed fluorescence or absence thereof (**Text SVII**). Up to 15 positive primary transformant seeds (T₁) were sown and treated with a heat cycle to increase Cas9 editing efficiency according to⁵³. Seeds from all T₁ plants that reached the generative phase were bulked per accession to generate T₂ seed pools.

From the T₂ seed pools, early flowering individuals were selected by screening more than 1000 plants per accession. Approximately 25 early flowering plants per accession were propagated to the T₃ generation (totaling more than 2,500 plants from more than 120,000 screened individuals). T₃ seeds were examined for presence of the transgene using the fluorescence reporter system. Flowering time was analyzed for one to six T₃ lines (five plants each) alongside five wild-type plants. One plant from one to two lines per accession of early flowering, non-segregating lines was selected for sequencing using Illumina 2x150 bp reads (see **Text SVII**).

Expression analysis

Three biological replicates per wildtype and one or two mutants were randomly distributed over three plates, grown at 20-22°C under long days (LD, 16 h light / 8 h dark), and sampled at ZT14 (two hours before dark). Several seedlings per line were pooled as one biological replicate. RNA from each biological replicate was extracted using a column-based protocol⁵⁴. 1 µg total RNA was reverse transcribed with RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, Waltham, USA)

using an oligo (dT) primer and cDNA was analyzed in a CFX384 Real-Time System Cyclor (BioRad, München, Germany). The relative quantification was calculated with the $\Delta\Delta C_t$ method using *ACT8* (AT1G49240) as a standard⁵⁵ and calibrated by biological replicate 1 (rep1) of Col-0.

Measurement of carbon isotope ratio

Mutant and wild types contrasts were grown in three pots representing three biological replicates at 20-22°C under LD. Rosettes were harvested at either the initiation of flowering or after 22 days after germination and tissue was dried at 60°C for 24 hours and homogenized. Dried material for 19 mutants and wild types was sent to Isolab GmbH (Schweitenkirchen, Germany) for an analysis of carbon isotope composition ($\delta^{13}\text{C}$) with ^{13}C -CF-IRMS. Four technical replicates per sample were analyzed. For a more detailed description of the procedure, see⁵⁶. Data are presented as $\delta^{13}\text{C}$ [‰] vs. V-PDB (**Fig. S25C**).

*Selection experiment with *flc* mutants*

50 mutant / wild-type contrasts, including a biological replicate, germinated and were grown in warm, non-water-limiting conditions in a growth chamber at 20-22°C long days. Plants were bottom watered 2 times a week. Flowering time was measured as days to first petal emergence and seed weight was measured as total weight of seeds harvested (mg). Values for flowering time and fitness were averaged across replicates. Given the conditions were comparable, instead of re-measuring $\delta^{13}\text{C}$ for the 50 wild-type accessions again, we assumed ranks of $\delta^{13}\text{C}$ across accessions are relatively constant across environments based on previous research²⁰, and imputed the data based on a larger $\delta^{13}\text{C}$ dataset of several hundred accessions²⁴ (NRMSE = 0.13; **Table S2**). For the *flc* mutants, we used the 29 values of $\delta^{13}\text{C}$ measured above, and then imputed the values for the remaining 21 accessions using those 29 values, the $\delta^{13}\text{C}$ data from²⁴ (**Fig. S26**), and the flowering time of the accessions from this experiment in order to ground the values predicted to this experiment and not general values of $\delta^{13}\text{C}$. Likewise, both the wild-type and mutant $\delta^{13}\text{C}$ values were centered by subtracting the mean $\delta^{13}\text{C}$ of their respective experiment (see Text SVIII). While replicating field-drought conditions in the growth chamber is challenging, this experiment generated substantial spread in fecundity fitness that enabled to detect natural selection (Fig. 3).

Data availability. The 1,862 phenotypes imputed for 1,135 wild *Arabidopsis thaliana* accessions and code to reproduce analyses and figures of this manuscript are available at github.com/moiexpositoalonso/Arabidopsis_Natural_Selection_Conflict. The genomic data from the 1001 Genomes of *Arabidopsis thaliana* are available at 1001genomes.org. The seed collection can be obtained at the Arabidopsis Biological Resource Center (ABRC) under the [ID CS78942](https://abrc.org/ID/CS78942). RNA-seq and WGS reads were deposited at ENA with the accession numbers PRJEB89960 and PRJEB89962. Seeds of *flc* lines are available upon request (Uli Lutz). Wild type progenitors are available under ABRC number [ID CS78942](https://abrc.org/ID/CS78942).

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